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## Module 1 Note: Network Theorems and Energy Properties of R, L, C

EE-AOC-581 (Mod 1 of 5, Sem 5 – Part I)

B.Tech (EE) (5th Sem)

07 Selected Students | Batch B2024–28

*Where the Story of the Exponential Family Begins*

*Add-On Course EE-AOC-581 / EE-AOC-681*

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**i Module Snapshot****⚡ Module:** 1 of 5, Semester 5 – Part I    **🕒 Duration:** 8 Contact Hours**📖 COs Addressed:** CO1, CO2    **📊 Bloom Levels:** BL1 (Remember), BL2 (Understand), BL4 (Analyse)**🏗️ Position in Course:** First module – lays the physical and mathematical foundation (Volt-second/Current-second balance, complex plane) that every later module refers back to, starting with Module 2's exponential family and reaching all the way to the FOC block diagram of Module 7**📖 Prerequisite Recall:** Basic Electrical Engineering, Network Theory (Semester 1–4)**1 🚩 Module Objective****🚩 Module Objective:** Reconnect students with network theory at a deeper physical level and establish Volt-second / Current-second balance as the engineering language used throughout the course. Introduce the complex plane as the natural home of EE mathematics.

Every topic in this module has already been seen once, in an earlier semester. The purpose here is not repetition. It is depth. Thevenin's theorem is no longer a circuit-reduction trick; it becomes the way an EV battery pack is modelled for a charger design. The impedance of an inductor is no longer a formula to substitute into; it becomes the reason a Buck converter inductor current ramps instead of jumping. This shift in depth is what the rest of the course depends on.

**2 🏗️ Network Theorems in Real-World Context****2.1 📡 Thevenin Equivalent – The Battery Model**

Any linear two-terminal network, however complicated internally, is seen by the outside world through only two numbers: an open-circuit voltage  $V_{th}$  and a series resistance  $R_{th}$ .

$V_{th}$  = open-circuit voltage at the terminals

$R_{th}$  = resistance seen at the terminals with all independent sources killed (voltage sources shorted, current sources opened)

**🔑 Why this matters beyond the textbook:** a 12 V lead-acid battery is not an ideal 12 V source. Under load it behaves exactly like a Thevenin equivalent: an ideal EMF  $V_{th}$  in series with an internal resistance  $R_{th}$  that grows as the battery discharges. The terminal voltage under a load current  $I_L$  is

$$V_{terminal} = V_{th} - I_L R_{th} \quad (1)$$

This single equation explains why a car battery's headlights dim when the starter motor draws a large  $I_L$ :  $R_{th}$  pulls the terminal voltage down.

**2.2 🏗️ Norton Equivalent and Source Transformation**

The dual representation replaces the series  $(V_{th}, R_{th})$  pair with a parallel current source and resistance  $(I_N, R_N)$ , related by

$$I_N = \frac{V_{th}}{R_{th}}, \quad R_N = R_{th} \quad (2)$$

Source transformation between the two forms is used constantly in later modules – for example, when a Buck converter's input stage is redrawn to simplify the small-signal model in Module 4.

## 2.3 Multi-Source Grid Context

In a distribution feeder with several distributed generation sources (solar inverters, a diesel genset, the grid connection itself), Thevenin reduction at the point of common coupling collapses the entire upstream network into one equivalent source. This is exactly how short-circuit level and fault current are estimated in a power system study, and it previews the NRPG-style multi-bus thinking used in transmission studies.

### Worked Example 1 – Battery Equivalent Circuit Reduction:

A network has an open-circuit terminal voltage of 12.6 V. When a 2  $\Omega$  load is connected, the terminal voltage drops to 11.8 V. Find  $R_{th}$  and predict the terminal voltage for a 5 A load.

*Solution.* Load current  $I_L = 11.8/2 = 5.9$  A. From  $V_{terminal} = V_{th} - I_L R_{th}$ :  $11.8 = 12.6 - 5.9 R_{th} \Rightarrow R_{th} = 0.1356 \Omega$ . For a 5 A load:  $V_{terminal} = 12.6 - 5(0.1356) = 11.92$  V.

## 3 Impedance of R, L, C versus Frequency

### 3.1 The Three Impedances

$$Z_R = R \quad (\text{no frequency dependence, no phase shift})$$

$$Z_L = j\omega L = \omega L \angle 90^\circ \quad (\text{current lags voltage by } 90^\circ)$$

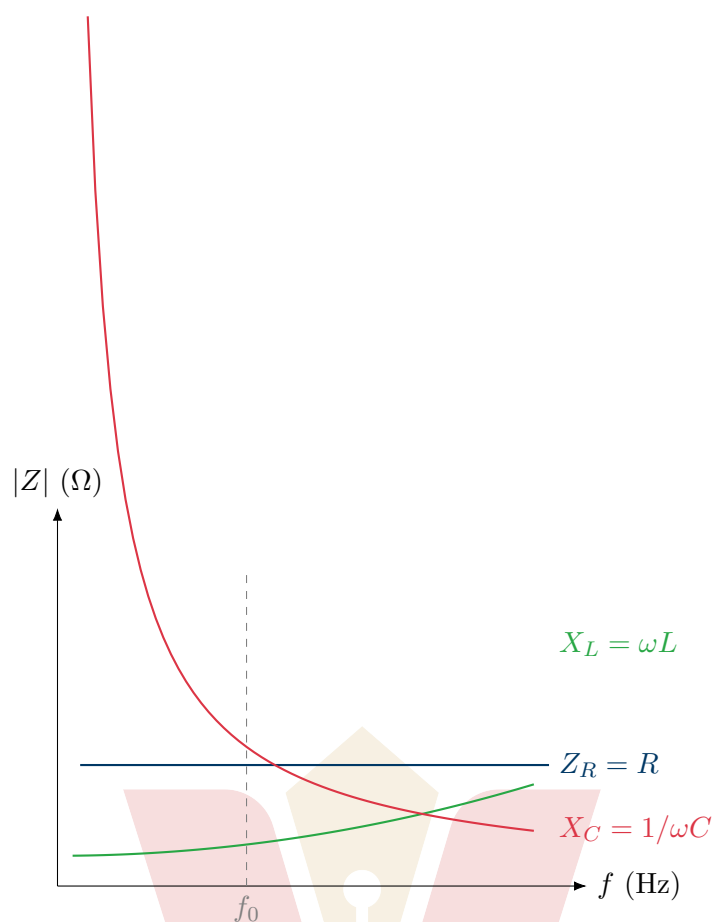
$$Z_C = \frac{1}{j\omega C} = \frac{1}{\omega C} \angle -90^\circ \quad (\text{current leads voltage by } 90^\circ)$$

| Element | $ Z $ at low $f$     | $ Z $ at high $f$    | Phase                |
|---------|----------------------|----------------------|----------------------|
| R       | $R$ (constant)       | $R$ (constant)       | $0^\circ$            |
| L       | $\rightarrow 0$      | $\rightarrow \infty$ | $+90^\circ$ (always) |
| C       | $\rightarrow \infty$ | $\rightarrow 0$      | $-90^\circ$ (always) |

Table 1: Limiting impedance behaviour of R, L, C

 **Physical reading:** an inductor looks like a short circuit at DC and an open circuit at very high frequency. A capacitor does the exact opposite. This single fact explains why an inductor is the natural energy-storage element for a Buck converter (it must pass DC load current while blocking switching ripple) and why a capacitor is the natural smoothing element at the output (it must pass switching ripple current to ground while holding the DC output voltage).

### 3.2 Reactance Curves



The crossover point  $f_0$ , where  $X_L = X_C$ , is the resonant frequency of the series LC combination – the same frequency that reappears as the LC double pole of a Buck converter's output filter in Module 4.

#### Worked Example 2 – Crossover Frequency:

For  $L = 220 \mu\text{H}$  and  $C = 470 \mu\text{F}$ , find the frequency at which  $X_L = X_C$ .

*Solution.* Setting  $\omega L = 1/(\omega C)$  gives  $\omega_0 = 1/\sqrt{LC}$ .  $\omega_0 = 1/\sqrt{(220 \times 10^{-6})(470 \times 10^{-6})} = 3111.5 \text{ rad/s}$ .  $f_0 = \omega_0/2\pi = 495.2 \text{ Hz}$ .

## 4 Volt-Second Balance of the Inductor

Starting from the defining relation of an inductor,

$$v_L(t) = L \frac{di_L}{dt} \quad (3)$$

integrate both sides over one complete period  $T$  of periodic steady-state operation:

$$\int_0^T v_L(t) dt = L[i_L(T) - i_L(0)] \quad (4)$$

In periodic steady state,  $i_L(T) = i_L(0)$  by definition (the waveform repeats), so the right-hand side is exactly zero. This gives the **Volt-second balance** condition:

$$\boxed{\int_0^T v_L(t) dt = 0} \quad (\text{steady state}) \quad (5)$$

**🔑 Physical meaning:** the net area under the inductor voltage waveform, positive area minus negative area, must be zero once the circuit has settled. Physically this is a statement about magnetic flux:  $\lambda = \int v_L dt = Li_L$ , and a periodic steady state requires the flux to return to its starting value every cycle. An inductor that violates this condition has current that ramps away indefinitely – it has not reached steady state, or the circuit will saturate the core.

**⚠️ Forward Reference:** the Buck converter relation  $V_o/V_{in} = D$ , derived formally in Module 4, is nothing more than Volt-second balance applied to the inductor across one switching period. This module is where that derivation becomes possible.

### 📦 Worked Example 3 – Volt-second Balance Applied:

An inductor in a switching circuit sees +10 V for 6  $\mu$ s and then  $-V_2$  for the remaining 4  $\mu$ s of a 10  $\mu$ s period. Find  $V_2$  for steady-state operation.

*Solution.* Volt-second balance:  $(10)(6 \times 10^{-6}) + (-V_2)(4 \times 10^{-6}) = 0 \Rightarrow V_2 = \frac{10 \times 6}{4} = 15$  V.

## 5 ⚖️ Current-Second Balance of the Capacitor

By the same argument applied to the dual element,

$$i_C(t) = C \frac{dv_C}{dt} \quad (6)$$

Integrating over one period  $T$  and using  $v_C(T) = v_C(0)$  in periodic steady state,

$$\boxed{\int_0^T i_C(t) dt = 0} \quad (\text{steady state}) \quad (7)$$

**🔑 Physical meaning:** the net charge delivered to a capacitor over one steady-state cycle is zero, since  $q = \int i_C dt = Cv_C$  and  $v_C$  must return to its starting value. Charge flows in during part of the cycle and an equal amount flows back out during the rest. This is precisely the mechanism by which an output capacitor absorbs high-frequency switching ripple current while its average, DC voltage stays fixed.

### 📦 Worked Example 4 – Current-second Balance Applied:

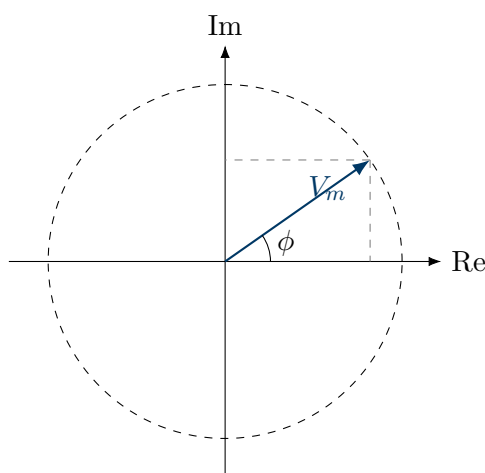
A capacitor in steady state receives a charging current of 2 A for 3 ms out of a 10 ms period. Find the discharging current magnitude for the remaining 7 ms, assuming it is constant.

*Solution.* Current-second balance:  $(2)(3 \times 10^{-3}) + (-I_2)(7 \times 10^{-3}) = 0 \Rightarrow I_2 = \frac{2 \times 3}{7} = 0.857$  A.

## 6 ∞ Euler's Formula and the Complex Plane

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (8)$$

This single identity is why every AC quantity in electrical engineering is written as a complex exponential. A phasor  $\tilde{V} = V_m \angle \phi$  is nothing but a **frozen snapshot** of the rotating quantity  $V_m e^{j(\omega t + \phi)}$  at  $t = 0$ : the projection of that rotating vector onto the real axis reproduces the familiar time-domain sinusoid  $V_m \cos(\omega t + \phi)$ .



🔑 **Why this matters:** once a signal is written as  $Ae^{st}$  with  $s = \sigma + j\omega$  a complex frequency, every network element's differential equation becomes an algebraic equation in  $s$  (this is the entire justification for using  $Z_L = j\omega L$  instead of solving  $v = L di/dt$  as a differential equation every time). Module 2 builds directly on this by showing that  $e^{-t/\tau}$ , the real, decaying member of this same exponential family, describes every RC and RL transient in the course.

## 7 ○ Impedance Locus and Introduction to Poles and Zeros

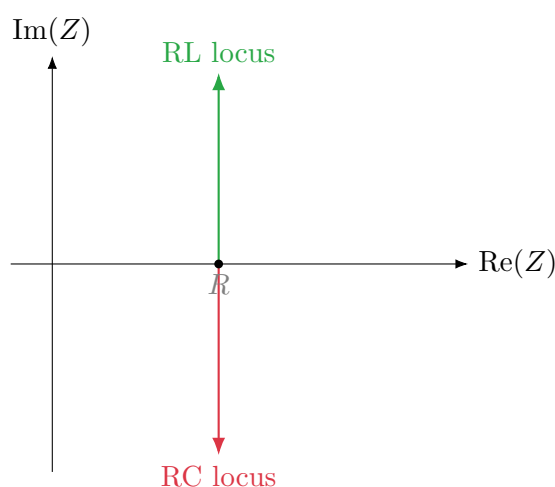
As frequency  $\omega$  is swept, the impedance  $Z(j\omega)$  of a network traces a path (a locus) on the complex plane. For a series RL circuit,

$$Z(j\omega) = R + j\omega L \quad (9)$$

the locus is a **vertical straight line** at  $\text{Re}(Z) = R$ , since the real part never changes while the imaginary part grows without bound. For a series RC circuit,

$$Z(j\omega) = R + \frac{1}{j\omega C} = R - j\frac{1}{\omega C} \quad (10)$$

the locus is again a vertical line at  $\text{Re}(Z) = R$ , this time extending downward (negative imaginary part) as  $\omega \rightarrow 0$  and approaching the real axis as  $\omega \rightarrow \infty$ .



🔑 **Bridge to later modules:** a network's transfer function  $H(s) = N(s)/D(s)$  has poles (roots of  $D(s)$ , where  $H(s) \rightarrow \infty$ ) and zeros (roots of  $N(s)$ , where  $H(s) \rightarrow 0$ ). The impedance

locus here is a first, concrete look at how a complex-plane picture encodes frequency behaviour – exactly the skill needed to read the Bode plots of Module 4 and the pole-zero placement discussion of Module 6’s compensator design.

## 8 Lab / Hands-On Session

** Lab / Hands-On:** Thevenin equivalent verification with real circuit and oscilloscope; Python/Scilab plots of impedance vs. frequency; phasor diagram and impedance locus Python visualisation

1. Build a two-source resistive network on breadboard. Measure  $V_{th}$  (open-circuit) and  $R_{th}$  (ohmmeter with sources killed). Verify by loading with a known resistor and comparing predicted vs. measured terminal voltage.
2. Using the accompanying Python script (`S01_rc_charging_multi_RC.py`), generate the RC charging overlay plot and the normalised universal curve. Import `S01_fig01_rc_charging_overlay.png` and `S02_fig02_normalized_curve.png` below once produced.
3. On the oscilloscope, capture the inductor voltage waveform of a simple chopper circuit and verify by planimetry (or by counting grid squares) that the positive and negative areas are approximately equal – Volt-second balance, observed directly.
4. Sweep frequency on a function generator across a series RC or RL network; record  $|Z|$  and phase at each frequency; plot the impedance locus by hand and compare with the theoretical vertical line derived above.

Figure 1: RC charging transient for the R, C grid of `S01_rc_charging_multi_RC.py` (insert after running the script; path relative to this note’s parent folder).

Figure 2: Universal, normalised RC charging curve – one shape for every R, C combination.

** Module 1 Assessment:** Lab notebook: circuit measurements and Python plots with physical interpretation paragraph for each result

**COs:** CO1, CO2    **Bloom Level:** BL1 (Re), BL2 (Un), BL4 (An)

## 9 Practice Problems

1. A network shows an open-circuit voltage of 9.0 V and a short-circuit current of 1.5 A at its terminals. Find  $R_{th}$ ,  $I_N$ , and the terminal voltage for a 4  $\Omega$  load.
2. A 100 mH inductor and a 47  $\mu\text{F}$  capacitor are placed in series. Find the frequency at which their reactances are equal.
3. An inductor voltage waveform is +20 V for 3 ms and  $-V_2$  for the remaining 2 ms of a 5 ms period. Find  $V_2$ .
4. Sketch the impedance locus of a series RLC circuit as  $\omega$  is swept from 0 to  $\infty$ , and identify the frequency at which the locus crosses the real axis.

5. Using Euler's formula, express  $v(t) = 20 \cos(314t + 30^\circ)$  as a phasor, and state the frequency in Hz.

## 10 💡 Key Takeaways

💡 **Key Takeaway:** Thevenin and Norton equivalents are not circuit-reduction exercises. They are how real sources – batteries, feeders, converters – are modelled at their terminals.

💡 **Key Takeaway:** Volt-second balance on L and current-second balance on C are the same statement (net flux/charge returns to its starting value each cycle) and they are the derivation engine behind every converter voltage-ratio formula in this course.

💡 **Key Takeaway:** The complex plane and Euler's formula are not extra mathematics bolted onto EE. They are the native language in which impedance, phasors, and later, transfer functions and Bode plots, are all written.

## 11 📖 Key References

1. C. K. Alexander and M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 6th ed., McGraw-Hill, 2016.
2. W. H. Hayt, J. E. Kemmerly, and S. M. Durbin, *Engineering Circuit Analysis*, 9th ed., McGraw-Hill, 2018.
3. R. W. Erickson and D. Maksimovic, *Fundamentals of Power Electronics*, 3rd ed., Springer, 2020. (Volt-second balance, Chapter 2)

## 12 👤 Note Prepared By

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 🏫 **Research Lab:** Tesla Research Lab (Departmental Research Lab, EE Dept., BCREC); Course venue: Sim Lab-III

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**End of: Module 1 Vivid Note – Network Theorems and Energy Properties of R, L, C**

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